EXPERIMENT NUMBER 6

Design and Evaluation of an Agentic RAG System using Adaptive Retrieval and Self-Correction using LlamaIndex

Name: Hitanshu Shah Roll Number / SAP: S069 / 60018240010

Q1. Difference Between Traditional RAG and Agentic RAG

Aspect	Traditional RAG	Agentic RAG
1. Retrieval	Retrieves relevant documents once for the given query.	Can perform multiple retrieval cycles when the initial evidence is insufficient.
2. Query Handling	Uses the user's original query for retrieval.	Can analyze and **reformulate the query** to find missing information.
3. Decision Making	Follows a mostly fixed sequence: retrieve and generate.	Makes decisions about whether to retrieve again, answer, or abstain.
4. Evidence Evaluation	Usually does not explicitly evaluate whether the retrieved evidence is complete.	Evaluates evidence for **relevance, completeness, and contradictions**.
5. Self-Correction	Has limited ability to correct retrieval or answer problems.	Can **self-correct** by changing the query and retrieving additional evidence.
6. Complex Queries	Works well for simple questions with directly available information.	Better suited for questions requiring information from **multiple documents or policies**.
7. Abstention	May generate an answer even when retrieved information is insufficient.	Can **abstain** and state that sufficient evidence was not found.
8. Groundedness	Generates an answer based on the retrieved context.	Can additionally check whether the generated answer is actually supported by the evidence.
9. Processing Cost	Generally requires fewer retrieval and reasoning steps.	May require additional retrieval and reasoning, resulting in higher latency and computational cost.
10. Workflow	**Query** → **Retrieve** → **Generate Answer**	**Query** → **Retrieve** → **Evaluate** → **Reformulate/Retrieve** → **Generate** → **Verify** → **Answer/Abstain**

Q2. Observation Table

Query	Traditional RAG Result	Agentic RAG Result	Retrieval Attempts	Relevant Evidence	Grounded	Latency
1. What is the minimum attendance required for ESE?	The minimum attendance required for the End Semester Examination (ESE) is **75 %**.	The minimum attendance required to be eligible for the End Semester Examination (ESE) is **75 %**.	1	Yes	Yes	4.44s
What topics are covered in Machine Learning?	The Machine Learning course covers the following topics: - Supervised Learning - Linear Regression - Logistic Regression - Decision Trees - Support Vector Machines - Clustering - Model Evaluation	The Machine Learning course covers the following topics: - Supervised Learning - Linear Regression - Logistic Regression - Decision Trees - Support Vector Machines - Clustering - Model Evaluation	1	Yes	Yes	5.08s
In which semester is Agentic AI offered?	Agentic Artificial Intelligence is offered in Semester VII.	Agentic Artificial Intelligence is offered in **Semester VII**.	1	Yes	Yes	4.38s
What should an internship report contain?	The internship report should contain the following sections: - Organization profile - Objectives - Work performed - Technologies used - Learning outcomes - Conclusion	An internship report should include: - **Organization profile** - **Objectives of the internship** - **Work performed** - **Technologies used** - **Learning outcomes** - **Conclusion** These elements are specified in the internship guidelines.	1	Yes	Yes	3.99s
What are the stages expected in a major project?	The major project is expected to progress through the following stages: 1. **Problem Identification** – Choose a suitable problem to address. 2. **Literature Review** – Survey existing work related to the problem. 3. **Problem Definition** – Clearly define the scope and objectives of the project. 4. **Proposed Methodology** – Design the solution approach and plan the work. 5. **Implementation** – Build and develop the system or solution. 6. **Results and Evaluation** – Test, analyze, and evaluate the outcomes. 7. **Report Preparation** – Compile a report that includes the sections: - Introduction - Literature Review - Problem Definition - Proposed Methodology - Implementation - Results and Evaluation - Conclusion - References 8. **Presentation & Viva Voce** – Present the project, demonstrate the system, and defend the work before the evaluation panel.	**Stages expected in a major project (as defined in the provided guidelines)** 1. **Problem identification** – Choose a suitable problem to address. 2. **Literature review** – Survey existing work related to the problem. 3. **Problem definition** – Clearly state the problem and its scope. 4. **Proposed methodology / design** – Design the solution (the “proposed method”). 5. **Implementation** – Build or develop the system/solution. 6. **Results and evaluation** – Test the system and evaluate the outcomes. 7. **Conclusion** – Summarize findings and implications. These stages correspond to the activities listed in the Major Project Guidelines (“identify a suitable problem, conduct a literature review, design the proposed solution, implement the system and evaluate the results”) and are reflected in the report sections (Introduction, Literature Review, Problem Definition, Proposed Methodology, Implementation, Results and Evaluation, Conclusion).	1	Yes	Yes	5.48s

Query	Traditional RAG Result	Agentic RAG Result	Retrieval Attempts	Relevant Evidence	Grounded	Latency
Does approved medical leave automatically make a student eligible for ESE?	No. An approved medical■leave application by itself does **not** automatically confer eligibility to sit for the End Semester Examination. Eligibility also depends on meeting attendance requirements (or obtaining approved attendance condonation) and fulfilling other examination regulations.	No. An approved medical■leave application **does not automatically confer eligibility** for the End Semester Examination. Both the Medical Leave and Attendance Condonation Policy and the End Semester Examination Policy state that approval of medical leave “does not automatically guarantee eligibility” and “by itself does not constitute final examination eligibility.” Eligibility is determined only after any required attendance■condonation decision (and other examination requirements such as registration) is approved.	1	Yes	Yes	10.44s
A student has 72% attendance and approved medical leave. Can the student definitely appear for ESE?	No. Because the student’s attendance is below the required 75 percent, the approved medical leave alone does not make the student automatically eligible for the End Semester Examination. The student must apply for attendance condonation, and only after an approved condonation decision (and any other examination requirements being met) can the student be confirmed as eligible to appear for the ESE.	No. The evidence shows that: * The normal minimum attendance required to sit the End Semester Examination (ESE) is **75 %**. * A student with only **72 %** attendance therefore falls short of the requirement. * Although the student has an **approved medical leave**, the policies state that “an approved medical leave application by itself does not constitute final examination eligibility” and that “students whose attendance falls below the normal attendance requirement may apply for attendance condonation.” * Eligibility is granted only after a **condonation decision** is approved (and after the student completes examination registration). Because the student’s attendance is below 75 % and the medical leave alone does not guarantee eligibility, the student cannot **definitely** appear for the ESE until a condonation request is approved (and registration is completed).	1	Yes	Yes	20.59s
A student has 80% attendance but did not complete examination registration. Is attendance alone sufficient?	No. While the student meets the attendance requirement (80 % ≥ 75 %), the policies state that students must also complete examination registration by the announced deadline. Failure to register can prevent a student from appearing for the End Semester Examination even when all other academic requirements, such as attendance, are satisfied.	The policies state that a minimum attendance of 75 % is required for eligibility, and the student’s 80 % satisfies that condition. However, the **End Semester Examination Policy** also requires that “students must also complete examination registration within the announced deadline,” and “failure to complete examination registration may prevent a student from appearing for the examination even if other academic requirements are satisfied.” Therefore, attendance alone is **not sufficient**; the student must also register for the exam. The evidence does not mention any exception that would waive the registration requirement, so without registration the student remains ineligible despite meeting the attendance threshold.	1	Yes	Yes	19.59s
What is the minimum CGPA required for scholarship?	Insufficient evidence found in the provided documents.	Insufficient evidence found in the provided documents. The system could not find sufficient evidence to answer the question definitively. Limitation: The exact minimum CGPA requirement as defined in the relevant scholarship notification or policy.	3	Yes	No	25.41s

Query	Traditional RAG Result	Agentic RAG Result	Retrieval Attempts	Relevant Evidence	Grounded	Latency
What is the hostel fee?	Insufficient evidence found in the provided documents.	Insufficient evidence found in the provided documents. The system could not find sufficient evidence to answer the question definitively. Limitation: Details about the hostel fee amount, payment schedule, and any related conditions.	3	Yes	No	16.57s